

A close-up, slightly blurred photograph of a person's hands holding a white smartphone. The person is wearing a light-colored, short-sleeved shirt. The background is dark and out of focus.

Lab 2: Weibo Data Crawler

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Technology Solution:

WEIBO SEARCH / WEIBO CRAWLER

Scrapes posts based on search terms

Sample Strategy: Determining the keyword list" that ensures the data is valid and relevant -
Determining the time range

Scrapes posts based on Users

Sample Strategy: Sampling the right people you are interested - determining the time range

Weibo Search

<https://github.com/dataabc/weibo-search>

Weibo Crawler

<https://github.com/dataabc/weibo-crawler>

From "Toy Task" to Engineering Solution

Build a Proxy IP Pool

Automated Daily Crawling

Batch User ID Acquisition

Build a Proxy IP Pool (Infrastructure)

- The Challenge: Using a single local IP or hardcoded Cookie leads to immediate bans.
- The Solution:
- Source: Acquire proxy resources (Free lists or Paid services via Taobao/XiaoHongShu are cost-effective).
- Code Implementation:
 - Refactor: Move from Hardcoded Cookies/IPs $\rightarrow$ Dynamic Lookup Lists.
 - Strategy: Implement random rotation of IPs and Cookies for every request.

Automated Daily Crawling

- The Challenge: Manual execution is inefficient for long-term monitoring.
- The Solution:
 - Logic: Modify the script to automatically set Start_Date and End_Date to the current day (Incremental Crawling).
 - Deployment: Use Crontab (Mac/Linux) or Task Scheduler (Windows) to run the script daily (e.g., at 02:00 AM).

Batch User ID Acquisition

- The Challenge: Manually copying User IDs is impossible for large-scale analysis.
- The Solution:
 - Scripting: Write a sub-script to parse search_results.
 - Method: Automatically extract user_id from keyword search results or "Follower/Following" lists to build a massive Target User Database.